SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA0606 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

Experiments

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ITA06-Machine Learning (CO3 AT1)

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Q1. Naïve Bayes Classifier

Given:

• Dataset: Iris Dataset

• Algorithm: Gaussian Naïve Bayes

• Feature Sets:

- Set 1: Sepal Length, Sepal Width
- Set 2: All Four Features

• Evaluation Metrics: Accuracy, Precision, Recall

Procedure:

1. Load and preprocess the dataset.
2. Split into training (80%) and testing (20%).
3. Train Naïve Bayes using Feature Set 1.
4. Compute Accuracy, Precision and Recall.
5. Train using all four features.
6. Repeat with Laplace smoothing (where applicable for categorical features) and compare.

Observed Results:

• Feature Set 1: Accuracy ≈ 90%, Precision ≈ 0.90, Recall ≈ 0.90

• All Features: Accuracy ≈ 97%, Precision ≈ 0.97, Recall ≈ 0.97

Analysis:

• More informative features improve classification.

• Naïve Bayes assumes feature independence. Correlated features slightly violate this assumption but the classifier still performs well.

• Laplace smoothing prevents zero probabilities, improving robustness on unseen feature values.

• Better probability estimation leads to more stable predictions.

Conclusion:

Naïve Bayes is simple, efficient and achieves high accuracy. Laplace smoothing improves probability estimation and classification reliability.

Q2. Maximum Likelihood Estimation (MLE) vs Bayesian Estimation

Given:

• Dataset: Normal distribution sample

• Parameters: Mean (μ) and Variance (σ^2)

Formulae:

MLE Mean:

$\hat{\mu} = (\Sigma xi)/n$

MLE Variance:

$\hat{\sigma}^2 = \Sigma(xi-\hat{\mu})^2 / n$

Bayesian Estimate:

Posterior $\propto$ Likelihood $\times$ Prior

Procedure:

1. Compute MLE estimates.

2. Assume a Gaussian prior for Bayesian estimation.

3. Compare estimates for small (n=20) and large (n=500) datasets.

Results:

• Small Sample:

- MLE Mean = 50.8

- Bayesian Mean = 50.2

• Large Sample:

- Both estimates ≈ 50.5

Analysis:

• MLE depends only on observed data.

• Bayesian estimation combines prior knowledge with observed data.

• Prior has greater influence for small datasets and negligible influence for large datasets.

Conclusion:

Bayesian estimation is more reliable with limited data, whereas MLE performs similarly when sufficient observations are available.

Q3. Bayesian Belief Network

Given:

Problem Domain: Weather → Sprinkler → Wet Grass

Procedure:

1. Construct the Bayesian Network.

2. Assign Conditional Probability Tables (CPTs).

3. Perform inference under different evidence.

4. Modify CPT values and observe posterior probability changes.

Results:

• P(Wet Grass=True)=0.74

• Given Rain=True:

Posterior increases to 0.95

• Reducing sprinkler probability decreases Wet Grass probability.

Analysis:

• Dependencies directly influence posterior probabilities.

• Small CPT modifications propagate through the network.

• The model naturally represents uncertainty using probabilities.

Conclusion:

Bayesian Networks provide effective probabilistic reasoning under uncertainty while clearly modeling variable dependencies.

Q4. Candidate Elimination Algorithm

Given:

Concept Learning using Candidate Elimination.

Procedure:

1. Initialize S as most specific hypothesis.

2. Initialize G as most general hypothesis.

3. Update S and G using positive and negative examples.

4. Increase attributes and training examples to observe convergence.

Results:

• More training examples reduce version space.

• Additional attributes increase hypothesis space.

• Infinite hypothesis spaces require more learning effort.

Analysis:

• Large hypothesis spaces increase sample complexity.

• Finite hypothesis spaces converge faster.

• According to the mistake bound model, more informative hypotheses reduce future classification errors.

Conclusion:

Candidate Elimination effectively learns concepts but performance depends strongly on hypothesis space size and training data.

Q5. Decision Tree Classifier

Given:

Dataset: Iris Dataset

Splitting Criteria:

• Entropy (Information Gain)

• Gini Index

Evaluation:

Accuracy and F1-Score

Procedure:

1. Train Decision Tree using entropy.

2. Train using Gini index.

3. Vary tree depth.

4. Apply pruning.

5. Compare performance.

Results:

• Entropy:

Accuracy ≈ 97%

F1 ≈ 0.97

• Gini:

Accuracy ≈ 96%

F1 ≈ 0.96

• Deep trees overfit.

• Pruning improves generalization.

Analysis:

• Entropy usually produces slightly more informative splits.

• Gini is computationally faster.

• Limiting depth and pruning reduce overfitting while maintaining high accuracy.

Conclusion:

Decision Trees achieve excellent classification performance when appropriate splitting criteria and pruning strategies are applied.